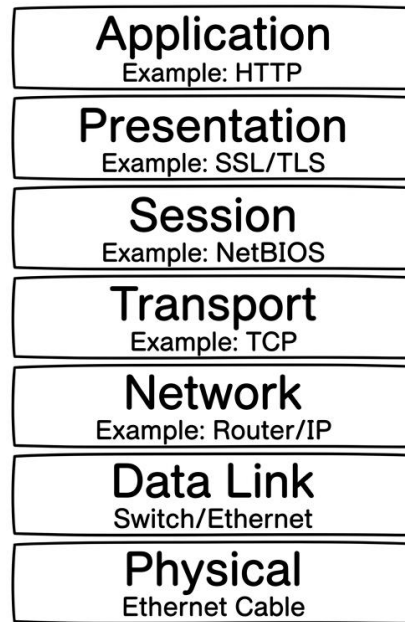


OSI Layer

Task 1: OSI 7-Layer Model



Task 2: OSI Layer Matching

- a) Encrypting data before sending: Presentation Layer (Layer 6)
- b) Routing packets across networks: Network Layer (Layer 3)
- c) Converting data to electrical signals: Physical Layer (Layer 1)
- d) Opening a web page in your browser: Application Layer (Layer 7)
- e) Detecting and correcting errors in frames: Data Link Layer (Layer 2)

OSI Layers in WhatsApp

When I send a message on WhatsApp, the Application Layer handles the app communication. The Transport Layer helps deliver the data reliably using protocols such as TCP. The Network Layer uses IP to route the data to the destination.

Other lower layers then send the data through Wi-Fi or mobile network to the receiver.

Task 4: Website Not Opening but Ping Works

The Network Layer is working because the IP address can be reached by ping. The problem may be at the Transport Layer if the required TCP port is blocked. It can also be at the Application Layer if the web service or HTTP/HTTPS is not working. So, I would check the Transport and Application layers first.